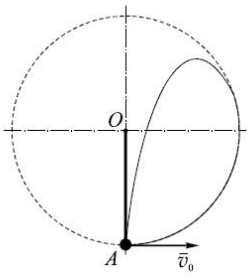
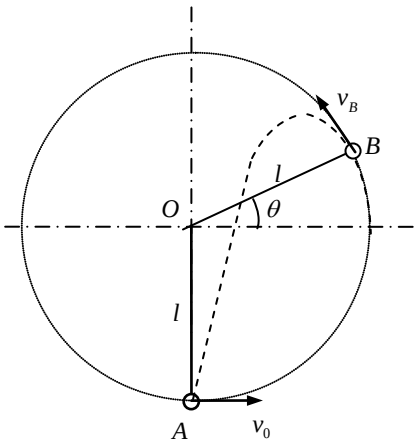
Problem D1-2009

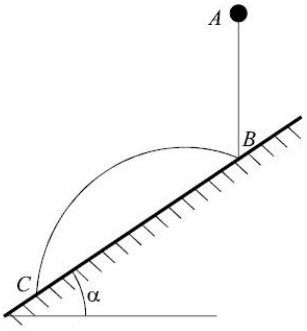


The load is hanged on a weightless thread with length l, the other end of which is fastened to the fixed point O. At start time the thread was in the lower vertical position OA and the load was given a certain initial velocity v_0. On further movement at a certain moment of time the thread got loose and the load began its free fall. Find at which initial velocity v_0 the thread will be tighten back just at the very moment when the load comes back to the origin A point.

质点 A 用长为 l、不计重量的细绳悬挂在固定点 O。初始时质点位于最低点，OA 垂直，给质点一初始速度 v_0，在后续运动的某瞬时绳子出现松弛，质点开始自由落体。若质点恰恰运动回到初始点 A 时，绳子正好拉紧，试求必须给质点的初始速度 v_0 是多少？

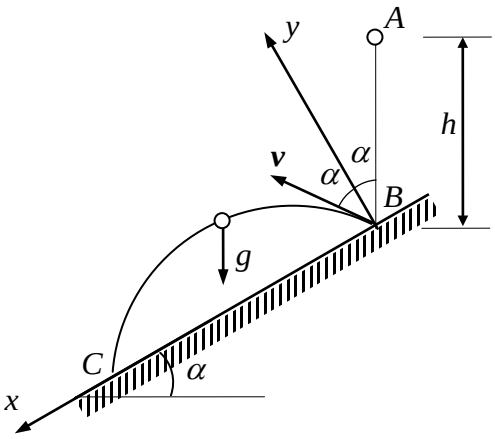


Problem D1-2011 (10 points)

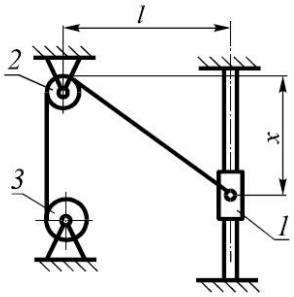


The ball falls down on the inclined plane with the horizontal angle α from the position *A* without initial velocity. Bouncing at point *B* it falls to such point *C* that *BC* = *l*. Considering the impact as absolutely elastic and ignoring air resistance define ball *AB*-section travelling time.

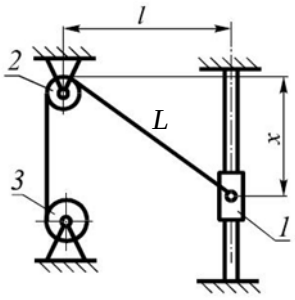
小球从点 *A* 无初速度下落到倾角为 α 的斜面上，落到斜面上 *B* 点，弹到点 *C*，*BC*=*l*。若碰撞是完全弹性碰撞，且忽略空气阻力的作用，求小球从 *A* 落到斜面 *B* 需要的时间。

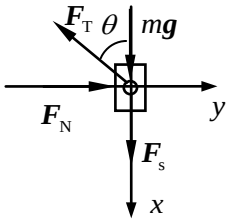


Problem D1–2014

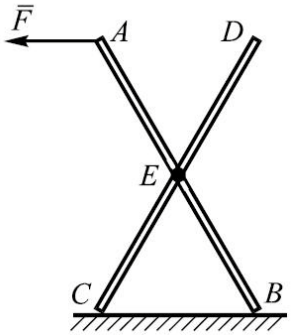


The load 1 of mass m rises along the vertical rod by the rope thrown over the block 2 (its sizes can be neglected). The block 2 is hanged at a distance l from the rod. The coefficient of friction between the load 1 and the rod is f. Determine the dependence of the rope tension force T on the distance x, if the rope is wounded on a uniformly rotating wheel 3 with a linear velocity v_0 .





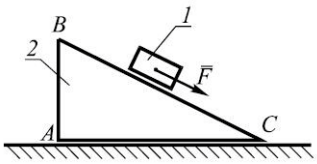
Problem D2-2009



The X-shaped construction consisted of two equal fixedly connected rods is placed on a smooth level floor. *AE=BE=BC=CE=ED*. At a certain moment of time force *F* was applied to *A* point. Find the force if it is known that at the moment when the force effect started the acceleration of *A* point was 4 times as high as the acceleration of *C* point. The mass of the construction is *m*.

X-型结构由两根相同的、固结在一起的杆组成，放置在光滑地面上，*AE=BE=BC=CE=ED*。结构的质量为 *m*。在某时刻力 *F* 作用在点 *A*。若已知在力 *F* 的作用下 *A* 点的加速度是 *C* 点的加速度的 4 倍，求力 *F*。

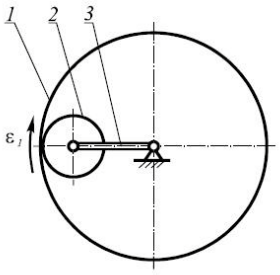
Problem D3-2009



Load 1 with weight m is sliding on side face prism 2 with weight $2m$ moving on the horizontal plane. Force F parallel to prism 2 BC face is applied to the load. There is no friction between the bodies. The prism may be made with different length ratios of its AB and AC edges. Find the maximum possible acceleration of prism 2.

质量为 m 的物块 1 在棱柱 2 的侧表面滑动，棱柱 2 质量为 $2m$，在水平面上运动。平行于棱柱 2 的 BC 面的力 F 施加于物块 1。各物体之间没有摩擦力。该棱柱可由其 AB 边和 AC 边的不同长度比制成。求棱柱 2 的最大加速度。

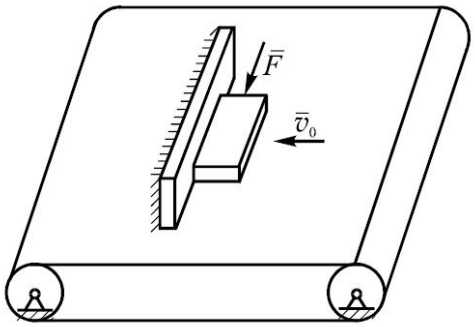
Problem D4-2009



Planetary gear train is placed in the vertical plane. The central wheel 1 with radius R is moving with a uniform angular acceleration ε_1. Wheel 2 – a homogeneous disk with radius r – is rolling skidless over the inner surface of central wheel 1. At some moment crankshaft 3 articulated with the centre of wheel 2 takes a horizontal position. At this moment the velocity of all points of the system is equal zero. Neglecting crankshaft weight find its angular velocity at the moment when it takes the vertical position for the first time.

行星轮系位于垂直平面内。半径为 R 的中心轮 1 以匀角加速度 ε_1 转动。半径为 r 的均质圆轮 2 在中心轮 1 的内表面作无滑动滚动。在某一时刻，曲柄 3 与轮 2 的中心成水平位置。此时系统所有点的速度均等于零。忽略曲柄的重量，当曲柄第一次取垂直位置时，求曲柄的角速度。

Problem D1 – 2010 (6 points)



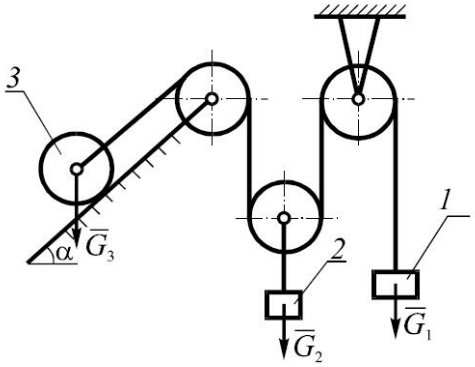
The load with the mass m on the conveyor belt is moving crosswise under the constant force F. The immovable barrier interferes with the lengthway movement of the load. The coefficient of friction between the load and the belt is f_1, between the load and the barrier is f_2. The speed of the belt is v_0.

1. Define the minimal force F_{min} when the motion is possible.

2. Find the maximum possible velocity of the load under the force $F = 2F_{min}$.

传送带上质量为 m 的载荷在恒力 F 作用下作横向运动。不可移动的挡板干扰了载荷的纵向运动。载荷与传送带之间的摩擦系数为 f_1，与挡板之间的摩擦系数为 f_2，传送带速度为 v_0。 1. 求能发生运动时的最小力 F_{min}。2. 求力 $F=2F_{min}$ 下载荷的最大可能速度。

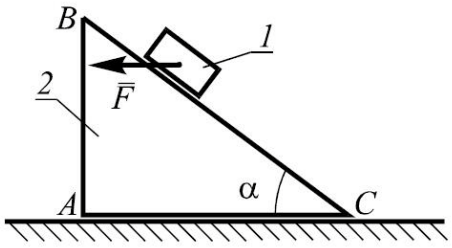
Problem D2 – 2010 (10points)



What is the weight of load G_1 in order that the load 1 is motionless in the mechanical system when $G_2 = 3G$, $G_3 = G$. Ignore the masses of unnumbered blocks, cables and friction in hinges. Consider the body 3 as a solid homogeneous disk. The coefficient of friction between the disk 3 and the surface is f. The angle α is given.

如图机构中，物块 B 重量为 $3G$，均质圆轮 C 重量为 G，圆轮 C 与斜面间的摩擦系数为 f，斜面倾角为 θ ，轮 1、2、3 及缆绳的质量均不计，铰接处摩擦不计，为使物块 A 处于静止，物块 A 的重量 G_A 应该为多少？

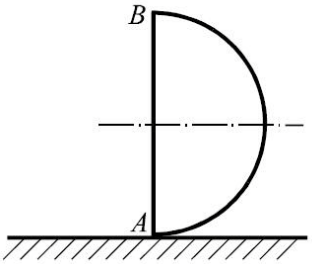
Problem D3 – 2010 (8 points)



The load 1 with the mass m_1 is sliding along the face of the prism 2 with the mass m_2 moving along the horizontal plane. The friction between the bodies is absent. The angle α is known. Find the minimal horizontal force F when the prism turn-over begins (i. e. the point C breaking-off from the surface).

Problem D3-2010 质量为 m_1 的物块 1 沿质量为 m_2 的棱柱表面滑动，棱柱沿水平面滑动，物体间的摩擦均不计，角 θ 已知。求使棱柱开始翻转（即：点 C 离开表面）的水平力 F 的最小值。

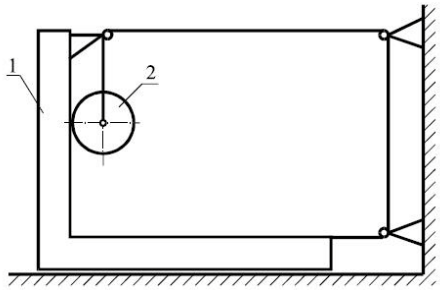
Problem D4 – 2010 (6 points)



Find the maximum angular velocity and maximum angular acceleration for the homogeneous semi-cylinder with the radius r if at the initial moment its AB side is vertical and the semi-cylinder is motionless. Consider the rolling motion is without slipping.

Problem D4-2010 如图所示，半径为 r 的均质半圆柱初始静止，直径 AB 铅直，在水平面纯滚动，求半圆柱的最大角速度和最大角加速度。

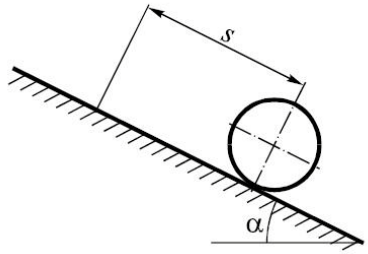
Problem D2-2011 (10 points)



Platform 1 with mass m_1 is riding on the smooth horizontal surface. The homogeneous disc 2 with mass m_2 is rolling on its vertical-way flatness. The platform and disc are connected by weightless rope-pulley system where there is no friction between the rope and construction units. Find the minimum value for the coefficient of friction between the disc and platform when the rolling occurs without slipping.

质量为 m_1 的平台 1 放在光滑水平面上，质量为 m_2 的均质圆盘 2 沿平台的铅垂面方向滚动。平台和圆盘由不计重力的绳索滑轮连接，忽略绳索及构件摩擦。若使圆盘能沿平台铅垂方向纯滚动，求圆盘与平台之间的摩擦系数最小值。

Problem D3-2011 (10 points)



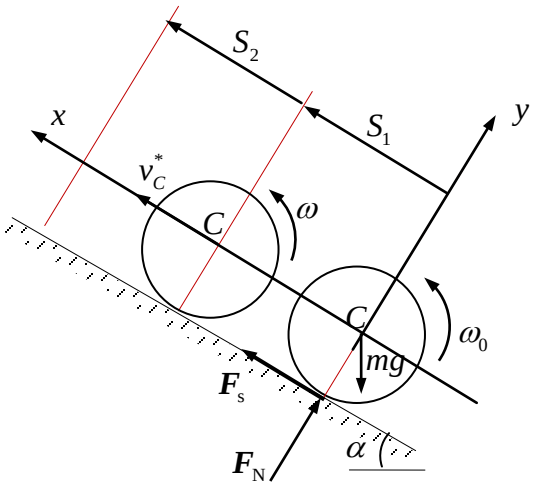
The solid homogeneous cylinder with radius r rotating with angular velocity ω_0 is placed on the inclined plane with horizontal angle α and it starts rolling uphill. Initial velocity of its center is zero. Coefficient of friction between the cylinder and plane is *f*. Define what maximum distance *s* the cylinder centre goes uphill along the inclined plane.

半径为 r 的实心均质圆柱具有初角速度ω_0，放在倾角为 α 的斜面上后往上滚动，圆柱中心的初速度为零。圆柱与斜面间的摩擦系数为 f。求圆柱沿斜面上行的最大距离 s。

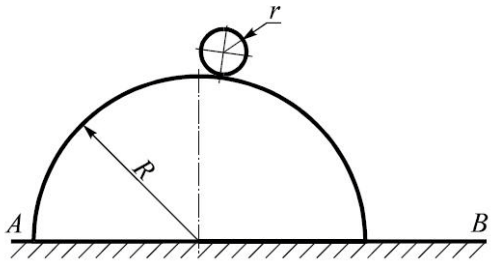
提示：

1.本题需要考虑两个阶段，阶段一：中心初速度为零、初角速度为ω_0 的圆柱放上斜面，开始时沿斜面既滚又滑地向上运动；阶段二：圆柱中心速度达到$v_C = r\omega$时，圆柱开始纯滚动，直到圆柱停止滚动；

2.圆柱沿斜面上行的距离为上述两个阶段行进的距离之和。



Problem D4-2011 (10 points)



The homogeneous cylinder with radius r starts rolling down from the top-point of the immovable semi-cylinder with radius $R = 6r$ with negligibly small initial velocity. The rolling of the cylinder is without any slipping. What velocity for the cylinder point which initial contact with AB plane will be immediately before the impact?

半径为 r 的均质圆柱从半径 $R = 6r$ 的固定半圆柱顶部开始向下纯滚动，初始速度忽略不计。在圆柱即将要接触 AB 平面的瞬间速度为多少？

提示：

1.本题需要考虑两个阶段，

阶段一：圆柱由静止开始在固定半圆柱上纯滚动至即将离开半圆柱表面阶段；

阶段二：圆柱离开半圆柱下落，直到接触 AB 平面。

2. 圆柱离开固定的半圆柱的条件是半圆柱的法向约束力为零；

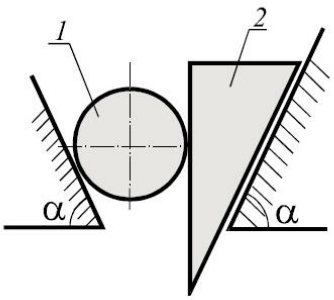
3. 用动能定理求圆柱即将离开半圆柱时质心的速度；

4. 用质心运动定理，求出离开时的角度；

5. 用动能定理求圆柱落至 *AB* 平面时圆柱中心的瞬时速度。

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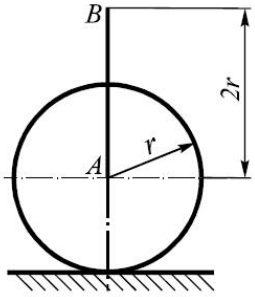
Problem D2-2012



Cylinder 1 with mass m_1 and wedge 2 with mass m_2 touching each other are moving between two inclined surfaces forming the identical angles α with the horizon. The frictions between the cylinder and the wedge as well as between the wedge and the surface are negligibly small. The cylinder is rolling along the inclined surface without slipping. Define what force the cylinder applies on the wedge.

质量为m_1的圆柱和质量为m_2 的楔形块相互接触，并沿着具有相同倾角α的斜面运动。圆柱与楔形块以及楔形块与斜面之间的摩擦均不计。圆柱沿着斜面纯滚动。求圆柱作用在楔形块上的力。

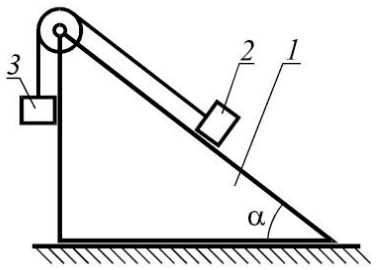
Problem D3-2012



The homogeneous disk with radius r is fastened with thin homogeneous rod AB. The disk and rod have the same masses. In the depicted position the quiescent system begins to move under the gravity forces. Ignoring the friction between disk and surface define the velocity of rod end B at the moment of the impact against the surface.

半径为 r 的均质圆盘与均质细杆 AB 固结。圆盘和杆具有相同的质量。在图示位置系统由静止在重力作用下开始运动。忽略圆盘和水平面的摩擦，求杆端 B 与水平面撞击瞬时，B 点的速度。

Problem D4-2012

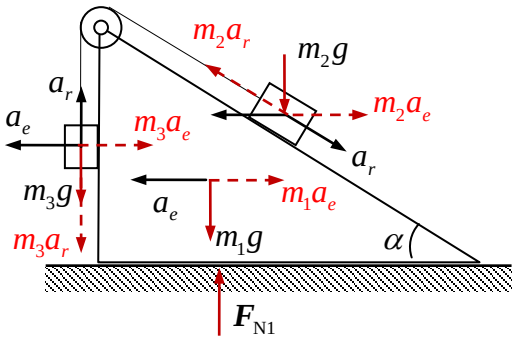


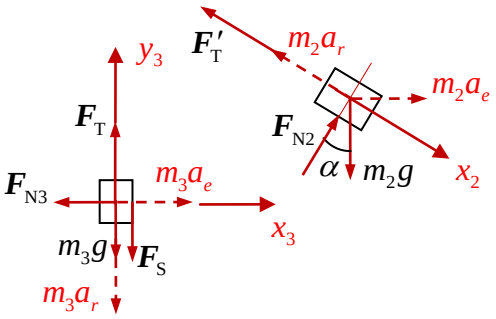
Prism 1 with mass m_1 may slide along the smooth horizontal surface. Two loads 2 and 3 with masses m_2 and m_3 ($m_2 \sin\alpha > m_3$) are connected by inextensible thread and sliding along the inclined and vertical prism faces respectively and the friction between load 2 and the prism is negligibly small. The coefficient of friction between load 3 and the prism face is f. Define the acceleration for prism 1. Block weight can be ignored. Angle α is known.

质量为 m_1 的棱柱 1，可沿光滑的水平面滑动。物块 2 和 3 质量分别是 m_2 和 m_3($m_2 \sin\alpha > m_3$)，用不可伸长的绳子连接，分别沿棱柱的斜面和铅垂面滑动，且物块 2 与棱柱斜面的摩擦不计，物块 3 与棱柱垂直表面的摩擦系数为 f。求棱柱 1 的加速度。滑轮的重量不计。角α已知。

提示：本题可用达朗贝尔原理求解！

1. **研究系统，虚加惯性力，列水平方向“平衡方程”；**
2. **研究物块 2，列 x_2 方向“平衡方程”；**
3. **研究物块 3，列 x_3 、y_3 方向“平衡方程”；**
4. **库仑定律 $F_S = fF_{N3}$**





D4 2012

系统: $\sum F_x = 0$.

$m_1 a_e + m_2 a_e + m_3 a_e - m_2 a_r \cos\alpha = 0$. ①

m_2 x_2: $m_2 a_e \cos\alpha + m_2 g \sin\alpha - m_2 a_r - F_T' = 0$. ②

m_3 y_3: $F_T - m_3 g - F_s - m_3 a_r = 0$ ③

x_3: $F_{N3} = m_3 a_e$,

$F_s = f F_{N3} = f m_3 a_e$ ④

④→③: $F_T = m_3 g + m_3 a_r + f m_3 a_e$ ⑤

⑤→②: $m_2 a_e \cos\alpha + m_2 g \sin\alpha - m_2 a_r$

$= m_3 g + m_3 a_r + f m_3 a_e$

$(m_2\cos\alpha - f m_3) a_e - (m_2 + m_3) a_r = m_3 g - m_2 g \sin\alpha$.

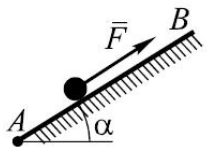
⑥: $a_e = \dfrac{m_2 \cos\alpha}{m_1 + m_2 + m_3} a_r$.

$(m_2\cos\alpha - f m_3) a_e - \dfrac{(m_2 + m_3)(m_1 + m_2 + m_3) a_e}{m_2 \cos\alpha} = m_3 g - m_2 g \sin\alpha$.

$\left[\dfrac{(m_2 + m_3)(m_1 + m_2 + m_3)}{m_2\cos\alpha} - (m_2\cos\alpha - f m_3)\right] a_e = m_2 g \sin\alpha - m_3 g$.

$a_e = \dfrac{m_2\cos\alpha (m_2 \sin\alpha - m_3)}{(m_2 + m_3)(m_1 + m_2 + m_3) - m_2\cos\alpha (m_2\cos\alpha - f m_3)} g$.

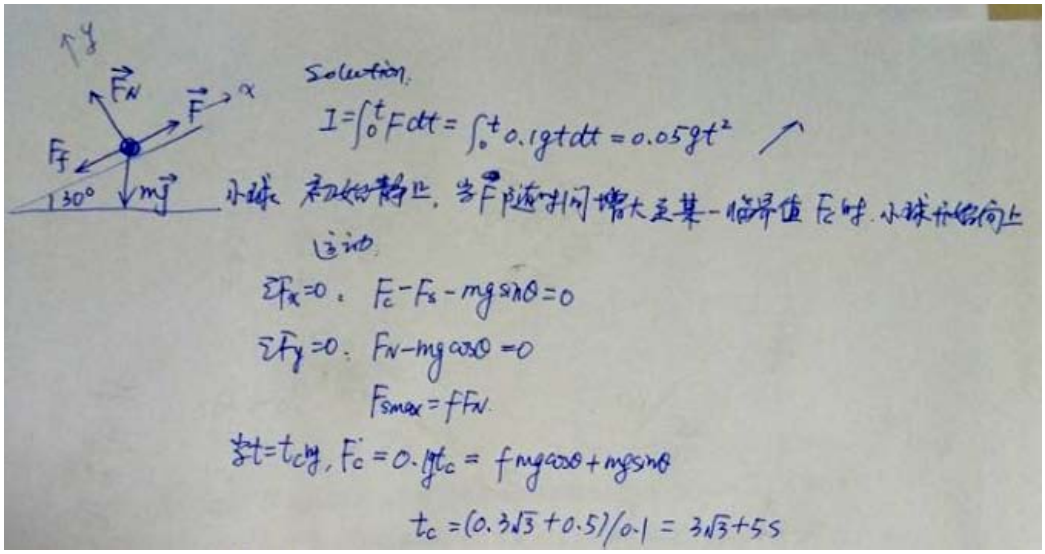
Problem D1–2013



At the initial time the material point of $\boldsymbol{m}$ = 1 kg is at rest on a rough surface. Coefficient of friction is equal to $\boldsymbol{f = 0.6}$. A force $\boldsymbol{F}$ = 0,1 $\boldsymbol{gt}$ (in Newtons) was applied to the material point.

Determine the dependence of the impulse of force and its work on time, if $\alpha = 30°$.

质点 m =1kg，初瞬时静止在粗糙的表面上，摩擦系数 f =0.6。力 F= 0.1gt(单位：牛顿)作用在质点上。若α=30°，求力的冲量及功与时间的关系。



当 $t \le t_c$ 时 $W_F = 0$

当 $t > t_c$ 时

$$W_F = \int_{t_c}^{t} F\,ds = \int_{t_c}^{t} F(t)\,v(t)\,dt$$

$$\sum F_x = ma_x \qquad F - F_s - mg\sin\theta = ma = m\frac{dv}{dt}$$

$$\int_{t_c}^{t} (F - F_s - mg\sin\theta)\,dt = \int_0^v m\,dv$$

$$\int_{t_c}^{t} (0.1gt - f mg\cos\theta - mg\sin\theta)\,dt = mv$$

$$0.1g \times \frac{1}{2}t^2 - (f mg\cos\theta + mg\sin\theta)t + \frac{1}{2} \times 0.1g t_c^2 = mv$$

$$v = 0.1g \times \frac{1}{2}t^2 - (0.3\sqrt{3} + 0.5)gt + \frac{1}{2} \times 0.1g t_c^2$$

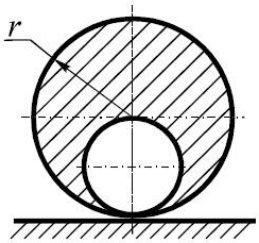
$$W_F = \int_{t_c}^{t} 0.1gt \times \left[0.1g \times \frac{1}{2}t^2 - (0.3\sqrt{3} + 0.5)gt + \frac{1}{2} \times 0.1g t_c^2\right] dt$$

$$= \frac{1}{2} \times (0.1g)^2 \times \frac{1}{4}t^4 - 0.1 \times (0.3\sqrt{3} + 0.5)g^2 \times \frac{1}{3}t^3 + \frac{1}{2} \times (0.1g)^2 t_c^2 \cdot \frac{1}{2}t^2 \Big|_{t_c}^{t}$$

$$= \frac{0.01}{8}g^2t^4 - \frac{0.01}{3} \times (3\sqrt{3} + 5)g^2t^3 + \frac{1}{4} \times 0.01 g^2 t_c^2 t^2$$

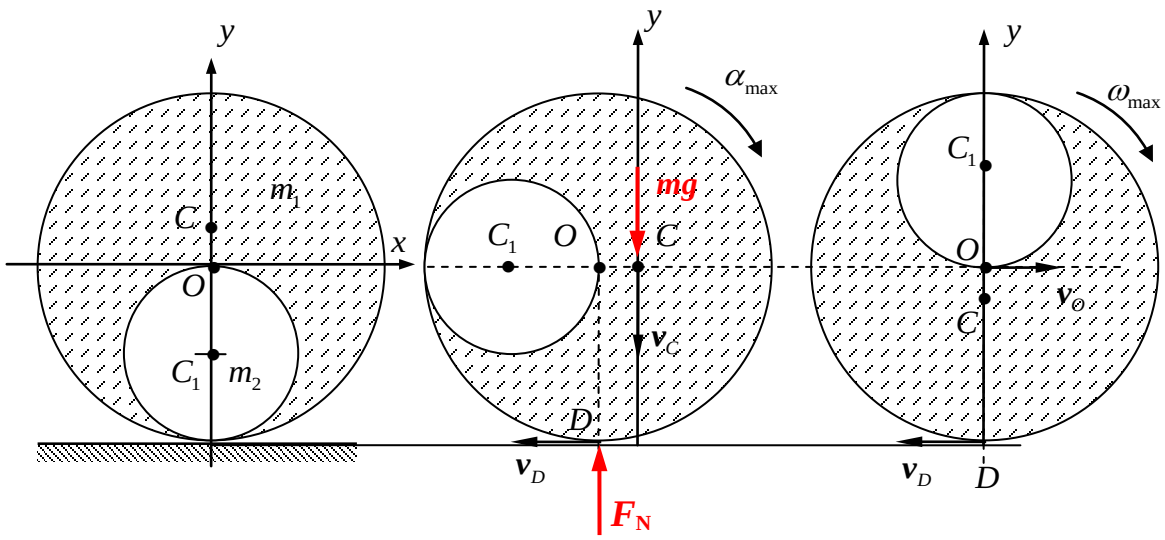
$$- \frac{0.01}{8}g^2 t_c^4 + \frac{0.01}{3} \times (3\sqrt{3} + 5)g^2 t_c^3 - \frac{1}{4} \times 0.01 g^2 t_c^4 - \frac{3}{8} \times 0.01 g^2 t_c^4$$

Problem D2–2013



A cylindrical hole with a radius ***r***/2 was cut from a homogeneous solid cylinder of radius ***r***. The resulting body was placed in a position of unstable equilibrium on a smooth horizontal surface, as shown in the figure. The body was given the infinitesimal angular velocity. Determine the largest angular velocity and angular acceleration of the body at its subsequent motion.

Problem D2-2013 如图所示，在半径为 ***r*** 的均质圆柱上切出一半径为 r/2 的圆柱孔后，放置在光滑的水平面上，处于非稳定平衡状态。给该物体一无限小的角速度，求后续运动中物体的最大角速度和角加速度。



(a) (b) (c)

提示： **1.求出物体的质心 C 位置；**

2.求出物体对质心 C 的转动惯量；

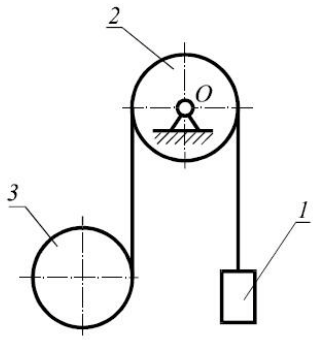
3.物体放在光滑面上，受力特点决定：质心 C 沿 y 轴方向运动！

4.图（b）处，物体的角加速度为最大；此时速度瞬心为点 O！点 O 的加速度沿 *x* 轴！

质心 C 的加速度沿 *y* 轴方向！

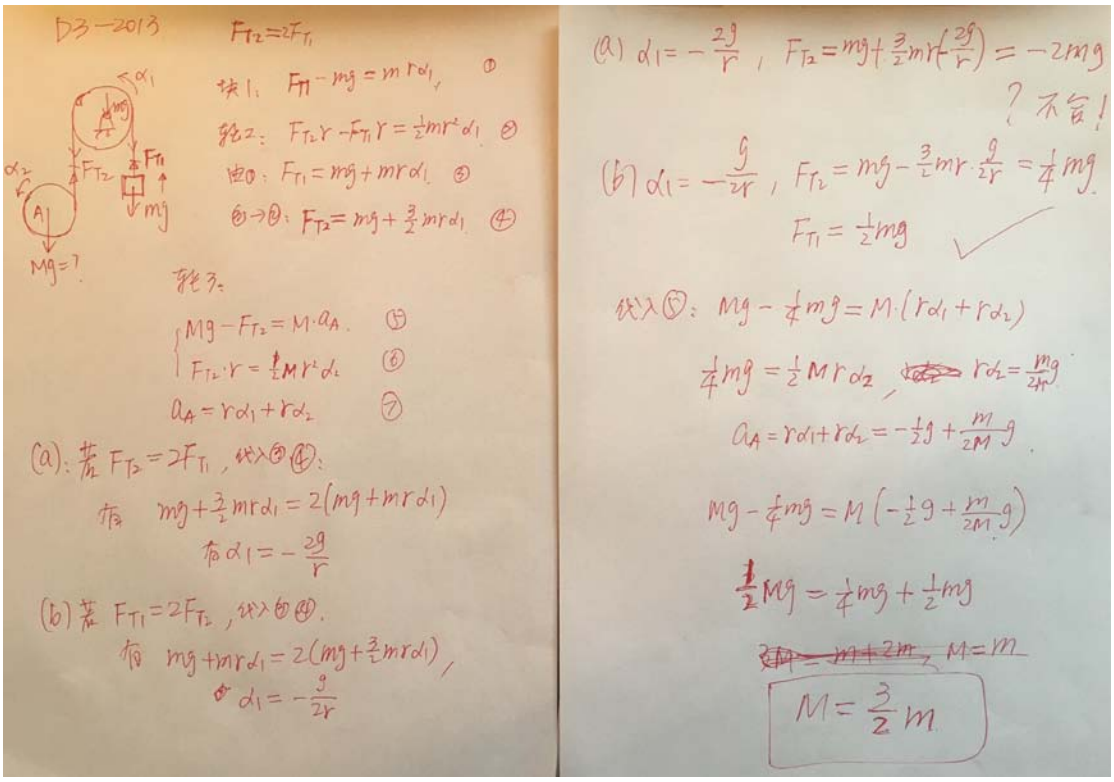
5.图（c）处质心位置最低，此时物体的角速度为最大！用积分形式动能定理求解！

Problem D3–2013

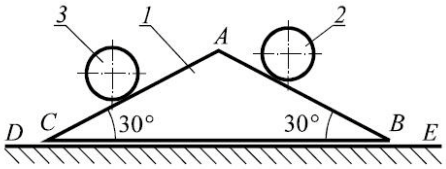


The load 1 is suspended from a weightless thread and over-thrown through the block 2, which rotates around a fixed horizontal axis ***O***. The other end of the thread is wound on disc 3. 1 and 2 bodies mass are the same and equal to ***m***. Block 2 and disk 3 are the solid homogeneous discs. The system starts from the rest under the action of gravity. Determine the values of body 3 weight when tension force in the left and right branches of the thread differs in 2 times. Friction forces are neglected.

重物 1 用自重不计、绕过滑轮 2 的绳子悬挂，滑轮 2 绕水平轴 O 转动，绳子的另一端缠绕在圆盘 3 上，滑轮 2 和圆盘 3 质量均为 *m*，均视为均质圆盘。系统在重力作用下由静止开始运动。当左右两边的绳子的张力相差 2 倍时，求圆盘 3 的重力。（摩擦不计）

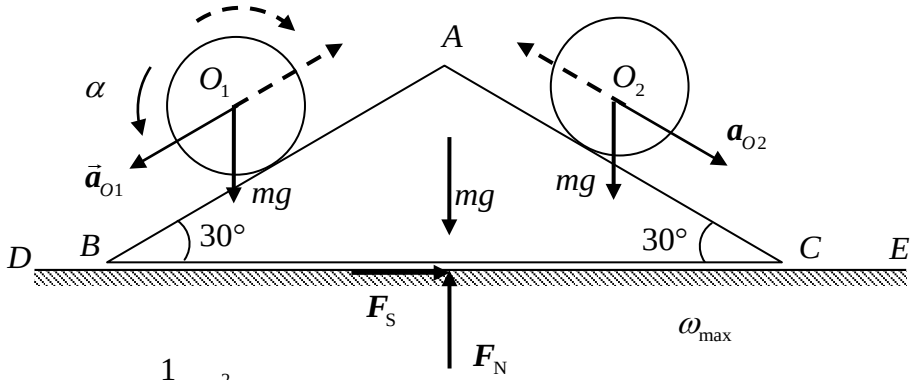


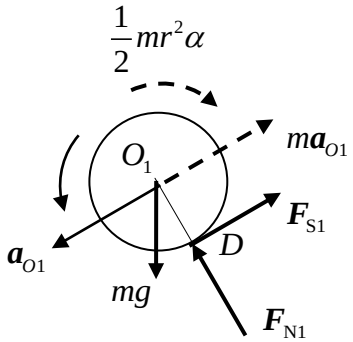
Problem D4–2013



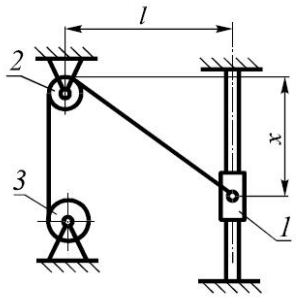
The triangular prism 1 with mass ***m*** is placed on the rough horizontal surface ***DE*** and can slide along this surface. A solid homogeneous cylinder 2 of the same mass ***m*** is put on the smooth face ***AB*** of the prism. A solid homogeneous cylinder 3 of the same mass ***m*** is put on the face ***AC*** of the prism and can roll along this face without slipping. Determine the values of the friction coefficient between the prism 1 and surface ***DE*** when the prism remains motionless despite the movement of cylinders.

质量为 m 的三棱柱放在粗糙的水平面 *DE* 上，并可沿该平面滑动。质量同为 *m* 的实心均质圆柱 2 放在棱柱的光滑的 *AB* 面上，质量同为 *m* 的实心均质圆柱 3 放在棱柱的 AC 面上，并沿该表面纯滚动。尽管两个圆柱运动，但三棱柱 1 保持静止，试求三棱柱 1 与表面 *DE* 之间的摩擦系数。





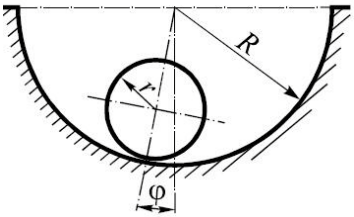
Problem D1–2014



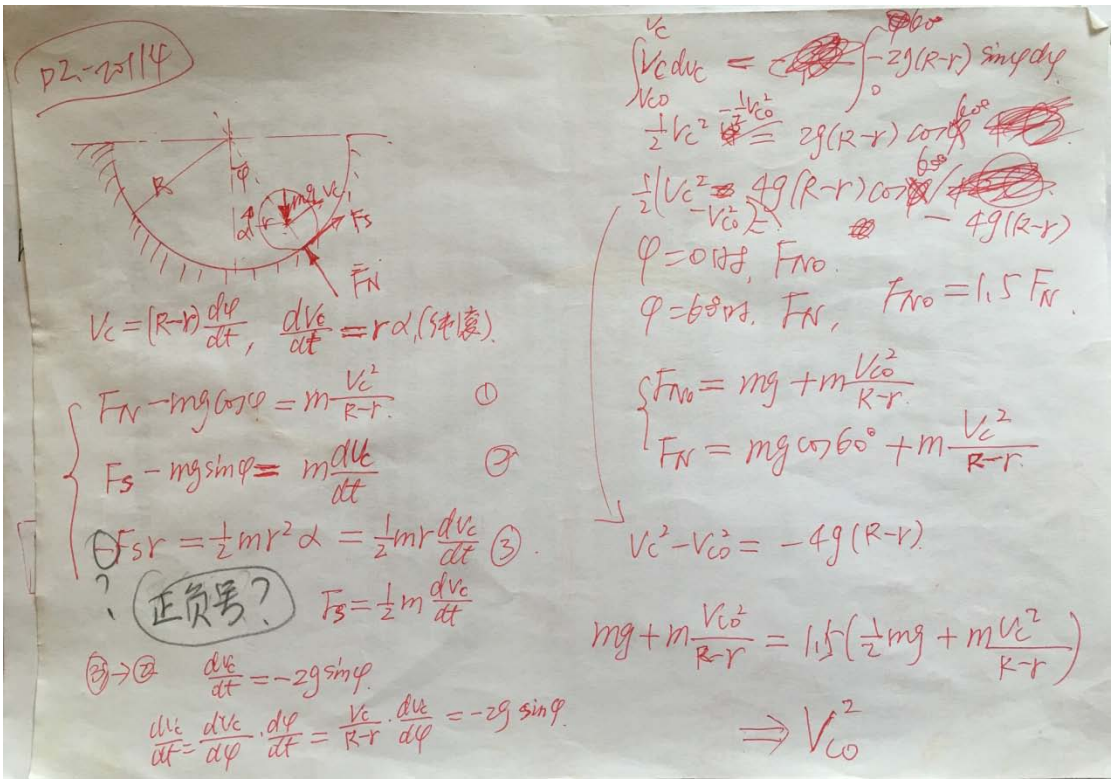
The load 1 of mass m rises along the vertical rod by the rope thrown over the block 2 (its sizes can be neglected). The block 2 is hanged at a distance l from the rod. The coefficient of friction between the load 1 and the rod is f. Determine the dependence of the rope tension force T on the distance x, if the rope is wounded on a uniformly rotating wheel 3 with a linear velocity v_0 .

滑块 1 质量为 m，被绕过滑轮 2 的绳子拉着沿铅垂杆上升（滑轮尺寸不计）。滑轮中心到杆的距离为 l，滑块 1 与杆之间的摩擦系数为 f。绳子绕在匀速转动的滑轮 3 上，速度为 v_0，试求绳子张力 T 与距离 x 的关系。

Problem D2–2014

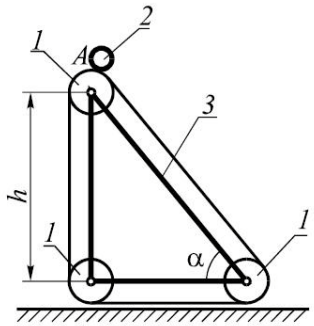


A solid homogeneous disk of radius r rolls without slipping along the arc of a circle of radius R. Find out the velocity which should be given to the center of the disk at its lower position for the decreasing of the normal cylinder pressure force in 1.5 times for the position of the disk when the angle $\varphi = 60\,^{\circ}$.



半径为 r 的均质圆盘沿半径为 R 的圆弧纯滚动。若在最低位置处圆盘所受的法向力是$\varphi=60°$处的 1.5 倍，试求在最低位置处圆盘中心应该具有的速度。

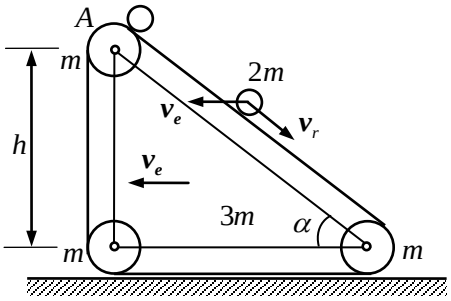
Problem D3–2014



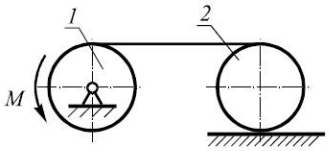
Each of three identical homogeneous cylinders 1 have mass m and radius r. The weightless inextensible belt is thrown over all cylinders (like a conveyor). The material point 2 of the mass $2m$ is located on a fixed conveyor in the highest point A, then the system starts to move. Body 2 does not slip on the belt and it comes off immediately after the contact with the lower cylinder at the right side of the system. Friction in the hinges and between the belt and the horizontal surface is negligible.

Determine the maximal velocity of the construction 3, if its mass is equal to $3m$. The distance h and the angle α are known.

三个质量为 m、半径为 r 的相同均质圆柱用不计自重、不可伸长的皮带围绕（类似于传送带）。质量为 $2m$ 的质点位于固定的传送带的最高点 A，随后系统开始运动。质点 2 在传送带上不打滑，且在接触系统最右边低处的圆柱瞬时立刻离开传送带。铰链处的摩擦及皮带与水平面间的摩擦均不计。求构件 3 的最大速度，若其质量为 $3m$，距离 h 及角度 α 已知。



Problem D4–2014

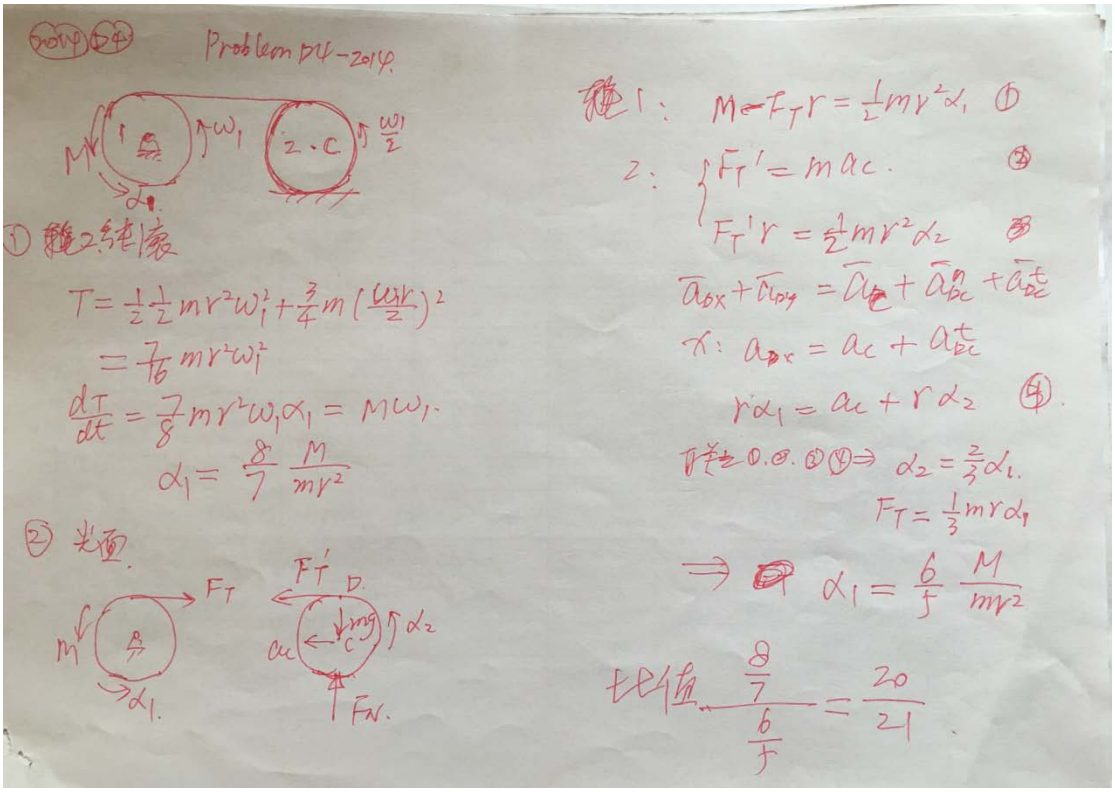


Homogeneous cylinder 1 rotates around a fixed axis under the action of pair of forces. It is connected with the same cylinder 2 with a non-stretchable thread. The cylinder 2 rolls on a horizontal surface.

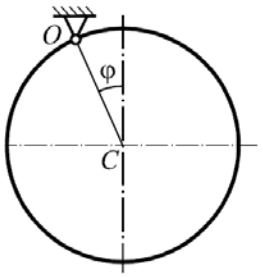
Two cases are possible: a) cylinder 2 is rolling without slipping; b) there is no friction between the cylinder 2 and the horizontal surface. Find out for which of these cases the value of angular acceleration of cylinder 1 is higher and how many times compared with the other case.

Problem D4-2014 均质圆柱 1 在力偶的作用下绕固定轴转动。圆柱 1 与圆柱 2 用不可伸长的绳子连接，圆柱 2 沿水平面滚动。圆柱 2 与圆柱 1 相同。

两种情况：a）圆柱 2 纯滚动；b）圆柱 2 与水平面之间无摩擦。求哪一种情况下圆柱 1 的角加速度更大，且是另一种的几倍。



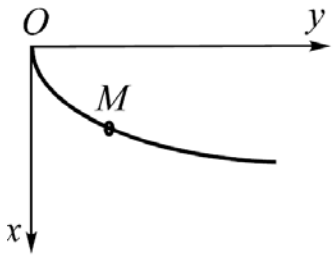
Problem D1–2015



Inhomogeneous disk oscillates in the vertical plane around the axis O. The density of the disk is proportional to the distance from the axis C, crossing the center of the disc. Determine how many times the periods of small oscillations differ for the case of the considered inhomogeneous disk and a homogeneous disk with identical radius and mass.

一非均质圆盘绕轴 O 在垂直的平面内振动，圆盘的密度和到圆心 C 的距离成正比，求相同条件下，非均质圆盘的微振动周期与同半径同质量的均质圆盘的微振动周期相差几倍？

Problem D2–2015



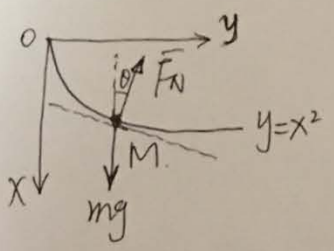
The ring M of mass m moves along the smooth wire of a parabola shape. The wire is located in a vertical plane. The ring starts to move from the top O of the parabola without initial velocity. Determine the maximum value of the interaction force between the ring and the wire if the wire shape is described by the equation $y = x^2$.

质量为 m 的圆环沿着抛物线状的光滑金属丝运动，金属丝位于垂直平面内，圆环无初速的从抛物线的顶点 O 处开始运动，若金属丝的形状可用方程 $y = x^2$ 表示，求圆环与金属丝之间相互作用力的最大值。

D2-2015

环M. 质量 m. O处静放.

曲线 $y=x^2$. 求 F_{Nmax}



解: 1. 动能定理:

$\frac{1}{2}mv^2=mgx$,

$\frac{1}{2}m(\dot{x}^2+\dot{y}^2)=mgx$ ①.

2. 牛顿定律:

$mg-F_N\cos\theta=m\ddot{x}$ ②

$F_N\sin\theta=m\ddot{y}$ ③.

3. 曲线 $y=x^2$

求导: $\dot{y}=2x\dot{x}$, ④

$\ddot{y}=2\dot{x}^2+2x\ddot{x}$ ⑤.

④式代入①式: $(1+4x^2)\dot{x}^2=2gx$, $\dot{x}^2=\frac{2gx}{1+4x^2}$ ⑥

求导: $8x\dot{x}^3+(1+4x^2)2\dot{x}\ddot{x}=2g\dot{x}$

$8x\dot{x}^2+(1+4x^2)\ddot{x}=2g$ ⑦

⑥式代入⑦式: $\frac{8gx^2}{1+4x^2}+(1+4x^2)\ddot{x}=2g$.

$\ddot{x}=\frac{1-4x^2}{(1+4x^2)^2}g$ ⑧

⑥、⑧代入⑤式: $\ddot{y}=\frac{2gx(3+4x^2)}{(1+4x^2)^2}$ ⑨.

由式②③得:

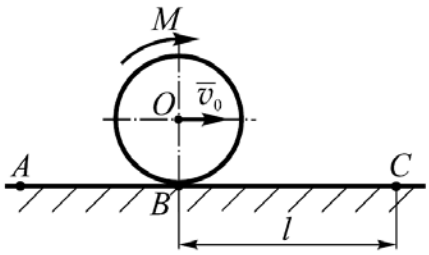
$F_N=m\sqrt{\ddot{y}^2+(g-\ddot{x})^2}$ ⑩

⑧、⑨式代入⑩式:

$$\left(\frac{F_N}{m}\right)^2=\left[\frac{2gx(3+4x^2)}{(1+4x^2)^2}\right]^2+\left[g-\frac{1-4x^2}{(1+4x^2)^2}g\right]^2$$

$\Rightarrow F_N=f(x)$.

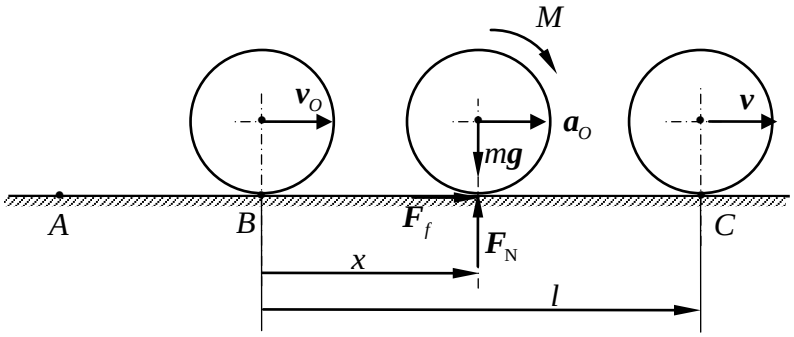
Problem D3–2015

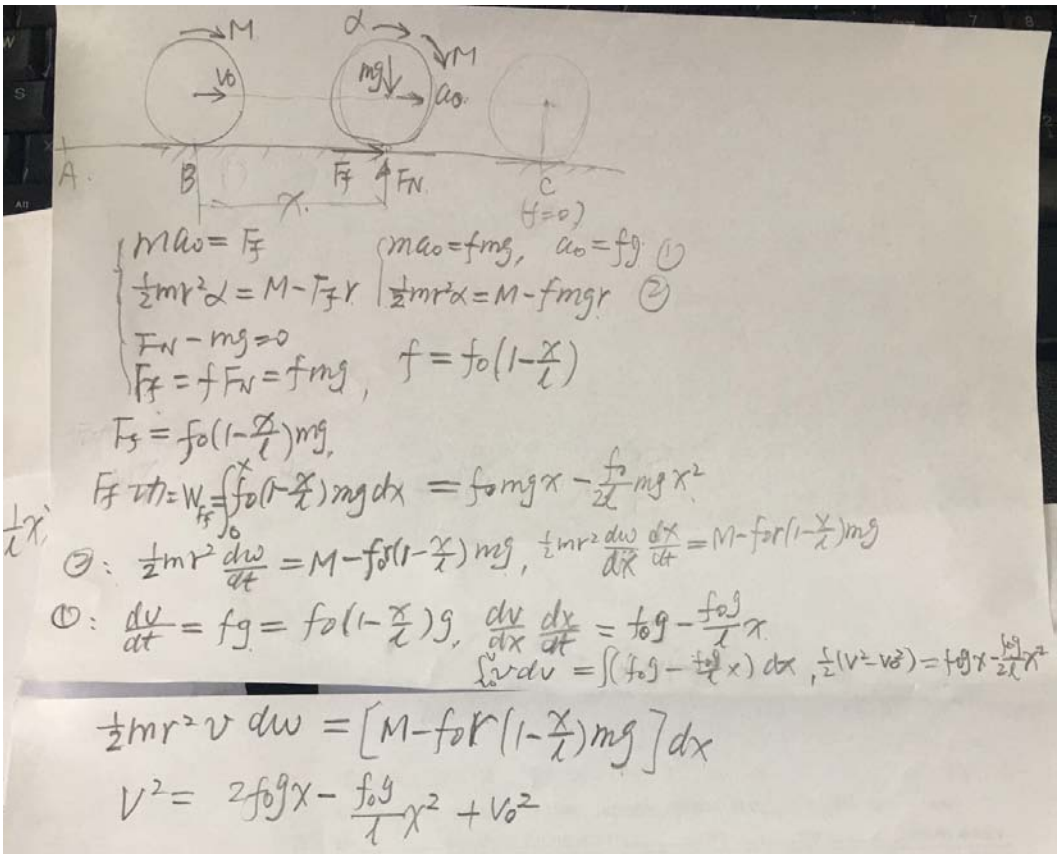


A solid homogeneous cylinder of mass m and radius r rolls along a horizontal plane under the influence of a constant torque M. The rolling of the cylinder in the distance interval AB is without slipping. When the cylinder is at the point B the velocity of the cylinder center O is equal to v_0. In the distance interval BC of length l, the friction coefficient between the cylinder and the plane decreases linearly from f_0 at point B to zero at point C. Determine the velocity of the cylinder center point O at position C.

A solid homogeneous cylinder of mass m and radius r rolls along a horizontal plane under the influence of a constant torque M, The rolling of the cylinder in the distance interval AB is without slipping. When the cylinder center O is equal to v_0. In the distance interval BC of length l, the friction coefficient between the cylinder and the plane decreases linearly from f_0 at point B to zero at point C. Determine the velocity of the cylinder center point O at position C.

质量为 m 半径为 r 的实心圆柱，在恒定的力矩 M 的作用下沿水平面滚动，AB 段做无滑动的滚动（纯滚动），到达 B 点时，柱心 O 的速度为 v_0，BC 段的距离为 l，圆柱与平面之间的摩擦系数线性减小。B 点处为 f_0，C 点处为零，求圆柱体运动到 C 点，圆柱中心点的速度。





Solution

$$\sum F_x = ma_O,\ F_f = ma_O = m\frac{\mathrm{d}v}{\mathrm{d}t},$$

$$\sum F_y = 0,\ F_{\mathrm{N}} - mg = 0$$

$$\sum M_O(\vec{\boldsymbol{F}}) = J_O\alpha\ ,\ M - F_f r = \frac{1}{2}mr^2\alpha,$$

$$F_f = fF_{\mathrm{N}}\ ,\quad f = f_0(1-\frac{x}{l})$$

Simultaneous the above equations, we get

$$F_f = f_0(1-\frac{x}{l})mg\ ,$$

$$f_0(1-\frac{x}{l})mg = ma_O\ ,\quad f_0g(1-\frac{x}{l}) = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}v}{\mathrm{d}x}\frac{\mathrm{d}x}{\mathrm{d}t} = v\frac{\mathrm{d}v}{\mathrm{d}x}$$

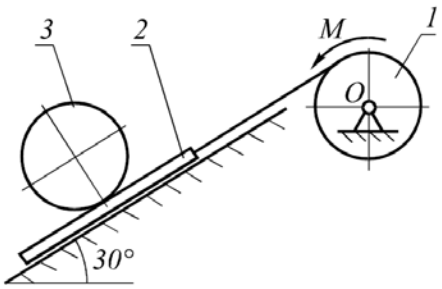
$$\int_0^x f_0g(1-\frac{x}{l})\mathrm{d}x = \int_{v_0}^{v} v\mathrm{d}v\ ,\quad v^2 = v_0^2 + 2f_0gx - \frac{f_0g}{l}x^2$$

When $x = l$ ， $v^2 = v_0^2 + 2f_0gl$,

hence $\underline{v = \sqrt{v_0^2 + 2f_0gl}}$

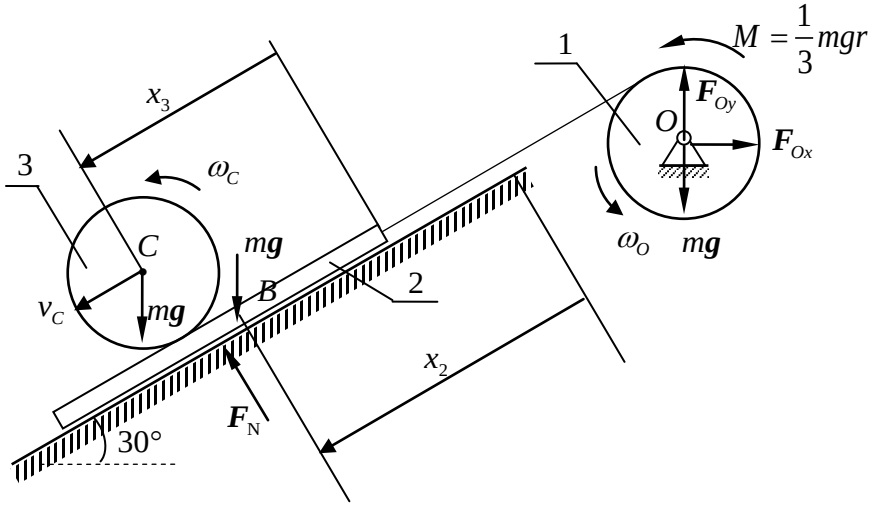
This is the velocity of the cylinder center point *O* at position *C*.

Problem D4–2015



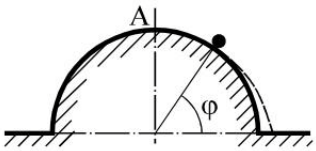
The mechanical system is located in a vertical plane and it consists of a homogeneous cylinder 1 connected with the desk 2. A homogeneous cylinder 3 rolls on the desk 2 surface without slipping. Weights of each body is equal to m. The radii of the cylinders 1 and 3 are also identical and are equal to r. Determine the acceleration of the cylinder 3 center of mass for the case when the body 1 is under the torque $M = \frac{mgr}{3}$. Friction between the desk and the inclined plane, as well as about the hinge O can be neglected.

机械系统位于垂直平面内，该系统由均质圆柱体 1 以及与之相连的桌面 2 组成，均质圆柱体 3 沿桌面无滑动的滚动，各物体质量均为 m，圆柱体 1 和 3 半径相同，均为 r，当圆柱体 1 在力矩 $M = \frac{mgr}{3}$ 作用下带动系统运动，求圆柱体 3 质心的加速度。假设桌面与倾斜面之间的摩擦，以及铰 O 处的摩擦都忽略不计。



Solution

Problem D1–2016



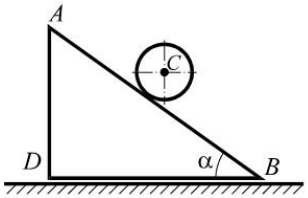
A small body, resting at the highest point *A* of a hemispherical convex surface, starts to slide down. The body can be considered as a material point.

1. Define the angle φ for the case of point coming off the surface if the surface is smooth.

2. Find the minimal value of the friction coefficient sufficient for the point movement along the surface without their disconnecting.

Problem D1-2016 质点 *A* 位于半球表面的最高点，由静止开始下滑。1.若表面光滑，求质点离开半球表面的角度φ； 2.求质点能不离开并沿半球表面运动的摩擦系数的最小值。

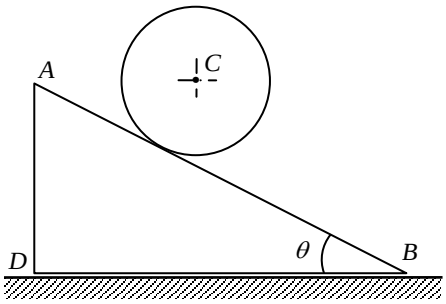
Problem D2–2016

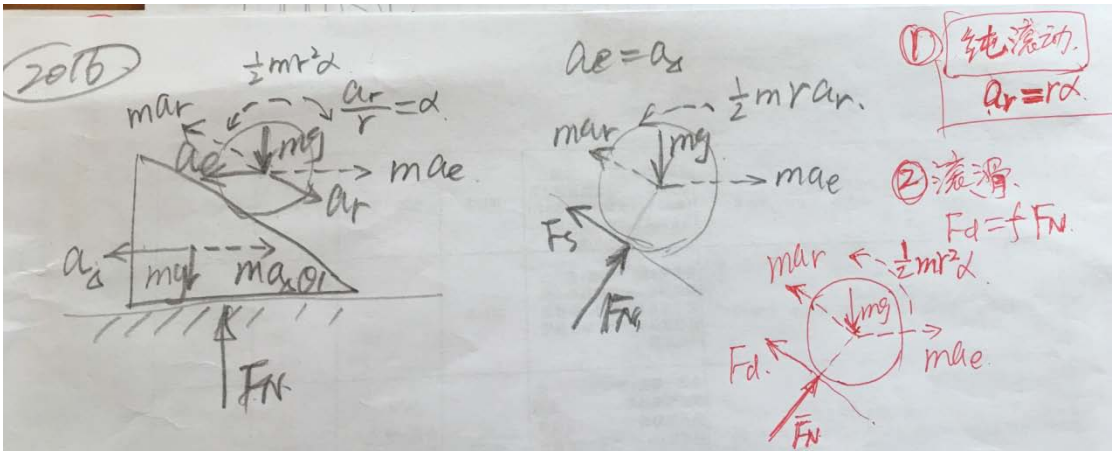


The triangular prism *ABD* of mass *m* is located on a smooth horizontal surface. $\alpha = 30°$. A solid homogeneous cylinder of the same mass *m* rolls along the edge *AB* of the prism. The friction coefficient between the prism and the cylinder is equal to *f*.

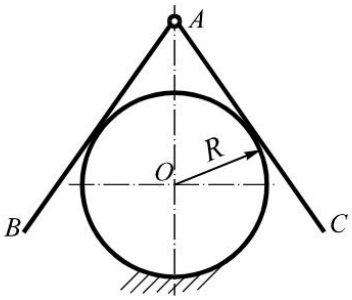
Determine the acceleration of prism if all bodies rested at the initial moment of time.

Problem D2-2016 质量为 *m* 的三棱柱 *ABD* 位于光滑的水平面上，α=30°，质量为 *m* 的均质圆柱沿三棱柱的 *AB* 边滚动。三棱柱和圆柱之间的摩擦系数为 *f* ，求系统静放瞬时，三棱柱的加速度。





Problem D3–2016



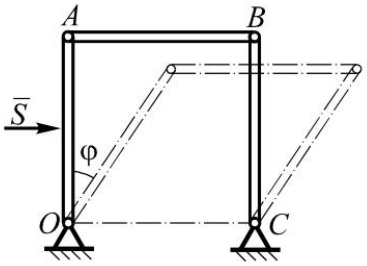
Two identical homogeneous rods AB и AC of length l ($3R < l < 5R$), are linked by hinge A and touch a motionless circle of radius R. Point O is the center of the circle. The system is located in the vertical plane. At the initial moment the velocities of bodies are equal to zero, the initial position of the rods is when $AO = 3R$.

Define the angular velocities and angular accelerations of the rods for the moment when $AO = 2R$. Friction forces are neglected.

Problem D3-2016 两根长均为 l、相同的均质杆 AB、AC，（$3R<l<5R$）铰接在 A 点并与半径为 R 的静止的圆 O 相接触，系统位于铅垂平面。初瞬时系统静止，杆位于 $AO=3R$ 的位置。求当 $AO=2R$ 时，两杆的角速度和角加速度。摩擦不计。

Problem D3-2016 两根长均为 l、相同的均质杆 AB、AC，（$3R<l<5R$）铰接在 A 点并与半径为 R 的静止的圆 O 相接触，系统位于铅垂平面。初瞬时系统静止，杆位于 $AO=3R$ 的位置。求当 $AO=2R$ 时，两杆的角速度和角加速度。摩擦不计。

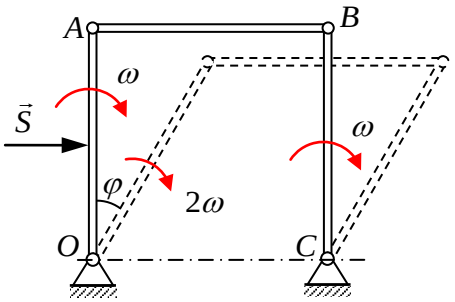
Problem D4–2016

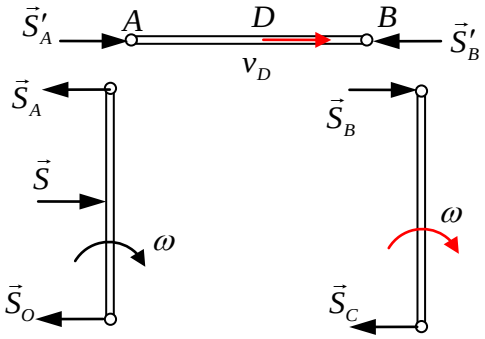


Three homogeneous rods of the same length and weight are pivotally connected as it is demonstrated in the figure. $AB = OC = l$. The rods were in a position of stable equilibrium in the vertical plane when the middle point of the left rod was hit by a horizontal shock impulse S.

Find out the value of the angle φ for the case when the rod OA angular velocity is 2 times more than its value directly after the impulse hit.

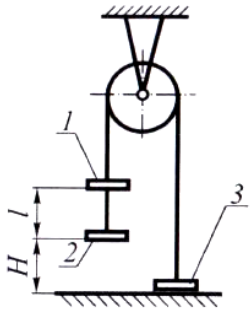
Problem D4-2016 三根等长、相同的均质杆铰接如图。$AB=OC=l$，杆系在铅垂面内处于稳定平衡时，受到水平冲量 S 作用。试求当 OA 杆的角速度为 2 倍冲击结束瞬时的角速度时，角 φ 的值。





Solution

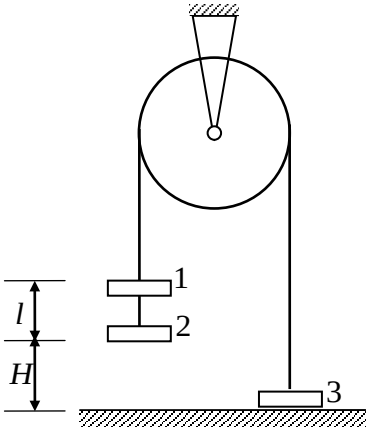
Problem D-1



The loads 1, 2, 3 are connected by a weightless inextensible thread thrown over a block of radius r and they have the same mass m. The dimensions of the loads can be neglected. The system starts to move without the initial velocity from the position shown in the figure. The dimensions H and l are given. At what maximal height will the load 3 rise, if the axial moment of inertia of the block is equal to mr^2, and the loads 1 and 2 remain stationary after reaching the floor?

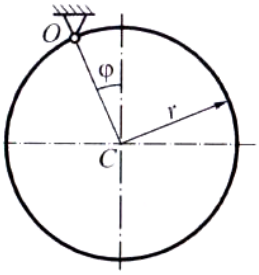
The load 1,2,3 are connected by a weightless inextensible thread thrown over a block of radius r and they have the same mass *m*. The dimensions of the loads can be neglected. The system starts to move without the initial velocity from the position shown in the figure. The dimensions *H* and *l* are given. At what maximal height will the load 3 rise, if the axial moment of the inertia of the block is equal to mr^2, and the loads 1 and 2 remain stationary after reaching the floor?

重物1,2,3质量均为*m*,用不计重量、不可伸长的跨过滑轮的绳子连接。重物的尺寸可以忽略。在如图所示位置，系统由静止开始运动。尺寸 *H* 和 *l* 已知。若滑轮对轴的转动惯量为 mr^2，重物 1 和 2 在运动到达地面后保持静止，重物 3 上升的最大高度是多少？



Solution:

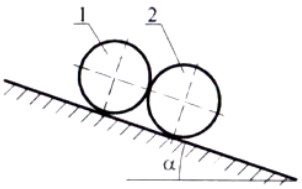
Problem D-2



A non-uniform disk of radius r oscillates in a vertical plane around the axis **O**. The density of the disk is proportional to the distance to the axis **C**, passing through the center of the disk. Determine the period of small oscillations of the disk.

半径为 r 的非均质圆盘绕轴 O 振动。圆盘的密度与到圆盘中心轴 C 的距离成正比。求圆盘作微幅振动的周期。

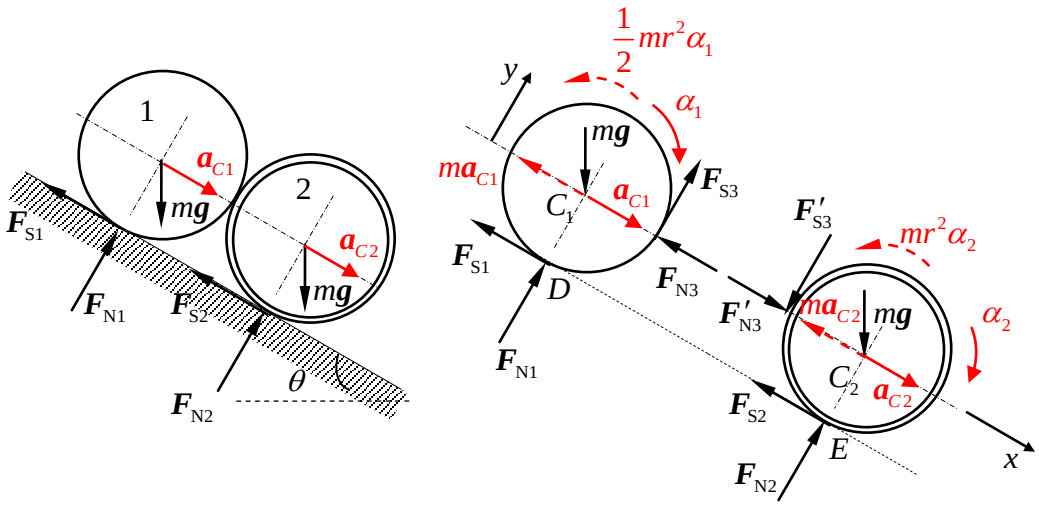
Problem D-3



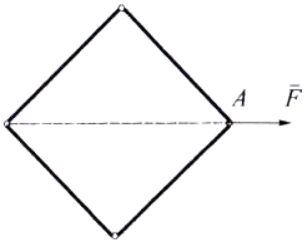
A rough uniform cylinder 1 and a thin pipe 2 of the same radii and masses, roll without slipping along a rough inclined plane forming an angle α with the horizon. The coefficient of sliding friction between the cylinders is *f*. Determine the acceleration of the centers of the cylinders of rolling bodies.

粗糙的均质圆柱 1 和与其质量、半径均相同的薄圆管 2，沿倾角为 θ 的斜面纯滚动。圆柱和圆管间的滑动摩擦系数为 f。求圆柱和圆管中心的加速度。

Solution:



Problem D-4



Four homogeneous rods of length l and mass m each are connected by hinges and form a square located on a smooth horizontal plane. The force F was applied to a hinge **A** in the shown direction. Before the force application the system rested. Determine the hinge **A** acceleration at the initial movement moment.

2019　四根长为 *l*、质量为 *m* 的均质杆铰接成一正方形，放置在光滑水平面上。力 *F* 沿图示方向作用在铰接点 *A* 上。在力作用之前系统静止。求铰链 *A* 在初始运动瞬时的加速度。

